DEVELOPER ONBOARDING & RUN GUIDE

GCMS Clinical AI Consultation Backend

Complete Setup, Download, Configuration, Seeding & Execution Manual

Project Name:	GCMS Clinical AI Consultation Backend
Target Audience:	New Backend Engineers, Full-Stack Developers, DevOps
Prerequisites:	Python 3.11+, Docker Desktop / Colima, Ollama (native macOS / Linux)
Repository URL:	https://github.com/HponeMyatThu/Clinic-ai.git
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INTERNAL ENGINEERING MANUAL: Follow these steps to get a fully working local development environment in under 10 minutes.

1. Prerequisites & Environment Setup

- **Operating System** — macOS (Apple Silicon M-series recommended for Metal GPU acceleration) or Ubuntu 22.04+.
- **Python 3.11 or 3.12** — Python 3.12 recommended (macOS: brew install python@3.12).
- **Docker & Docker Compose** — Required for running local PostgreSQL 16 (with pg_trgm) and Redis.
- **Ollama** — Native host installation for local AI inference (brew install ollama).

2. Step-by-Step Installation

Step 2.1: Clone and Create Virtual Environment

```
git clone https://github.com/HponeMyatThu/Clinic-ai.git gcms_ai_backend
cd gcms_ai_backend
python3.12 -m venv .venv
source .venv/bin/activate
```

Step 2.2: Install Project Dependencies

```
pip install -r requirements.txt
```

Step 2.3: Configure Environment Variables

```
cp .env.example .env
```

3. Starting Database & Local AI Services

Step 3.1: Start PostgreSQL & Redis via Docker

```
docker compose up -d
# Verify containers are running healthy:
docker compose ps
```

Step 3.2: Start Local Ollama and Pull Reasoning Model

```
# Start daemon in a separate terminal:
ollama serve

# Pull the recommended 8B reasoning model:
ollama pull qwen3:8b
# (Alternative model: ollama pull llama3.1:8b)
```

4. Database Seeding & Schema Verification

Populate the database with 50 primary care SNOMED CT concepts and 40 Singapore Drug Dictionary (SDD) medications:

```
python -m scripts.seed_dev_data
```

[SUCCESS] Verification Command in Docker

docker exec -it gcms-postgres psql -U gcms -d gcms_ai -c "SELECT 'snomed_concepts', count(*) FROM snomed_concepts;" Should return 50 concepts and 196 descriptions.

5. Launching Server & Testing APIs

```
uvicorn app.main:app --reload --port 8000
```

- **Interactive Swagger UI** — http://localhost:8000/docs
- **ReDoc API Documentation** — http://localhost:8000/redoc
- **Readiness Health Probe** — http://localhost:8000/api/v1/health

6. Common Troubleshooting

- **relation 'snomed_descriptions' does not exist** — Database not seeded. Run 'python -m scripts.seed_dev_data'.
- **Connection refused on port 5432** — Docker stopped. Run 'docker compose up -d'.
- **Ollama unreachable in /health** — Daemon not running. Run 'ollama serve' in a terminal.
- **ModuleNotFoundError** — Virtual environment not activated. Run 'source .venv/bin/activate'.